

20/08/26

Signal Processing

◦ Recap: Time-invariant systems, examples ✓

◦ Causal, stable systems, examples

◦ Linear time-invariant (LTI) systems.

- Convolution sum

- Runge averaging, differencing systems

◦ A system is said to satisfy a property iff the property holds for all input

◦ Time-invariant systems: A system $\mathcal{T}\{\}$ is said to be time-invariant if the following holds for all n_0 . If $y[n] = \mathcal{T}\{x[n]\}$ $x_1[n] = x[n - n_0]$,

then the system is time invariant if $\mathcal{T}\{x_1[n]\} = \mathcal{T}\{x[n - n_0]\} = y[n - n_0]$

◦ Example: $y[n] = \sum_{k=-\infty}^n x[k]$

◦ Example: $y[n] = x[Mn]$, M is a positive integer. ($M > 1$)

Aside: compressor or "down sampler". Note that this system plays an important role in wavelets / multi-rate signal processing.

◦ Systems like averaging / differencing are time-invariant systems.

- Let $y[n] = \mathcal{T}\{x[n]\} = x[Mn]$.

- ① let $M=2$

- $x_1[n] = x[n - n_0]$ (for an arbitrary n_0)

- $y_1[n] = \mathcal{T}\{x_1[n]\} = x_1[Mn] = x_1[\underbrace{n}_{= Mn}] = x[\underbrace{n' - n_0}_{\text{delayed version of } x}]$
(from ①) $= Mn = x[Mn - n_0]$ - ②

- $y[n - n_0] = x[M(n - n_0)]$ - ③

($\because y[0] = x[2 \cdot 0]$, $y[1] = x[2 \cdot 1]$ - - - $y[k] = x[2 \cdot k]$)

Reminders

- Office hours on Thu @ 2pm in EE605

- Text book: Discrete Time Signal Processing by Oppenheim & Schaffer. 2nd Edition. Prentice Hall.

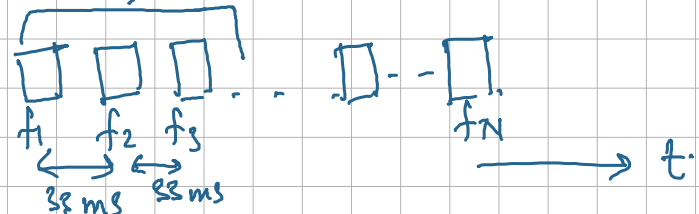
- $y_1[n] = x[Mn - n_0] \neq (y[n - n_0] = x[M(n - n_0)])$. from
② and ③

$\Rightarrow$ Compressor system is not time-invariant.

o Q: Have you come across systems where the output at the current time n i.e., $y[n]$ depends on a future input say $x[n+k]$? $k > 0$

Ex: $y[n] = x[n+1] - x[n-1]$?

Ex: Video compression.



The diagram shows a sequence of video frames represented by boxes labeled $f_1, f_2, f_3, \dots, f_N$ along a horizontal axis labeled t . Double-headed arrows between f_1 and f_2 , and between f_2 and f_3 , are labeled "33ms", indicating the time interval between consecutive frames.

o f_1 is encoded like an image.

o However, f_2 is encoded not like an image, but like a change with respect to f_1 . This strategy exploits the correlation present in the video sequence. This is a causal approach

o In bidirectional coding, compression efficiency is increased by using both f_1 and f_3 to encode f_2 . This means that we look into the future to encode the present.

$\Rightarrow$ A non-causal system. Again, since video frames are highly correlated in a small temporal neighborhood.

* Real life example: The warning system in the Golconda fort is a causal system. *

Definition: A system $T\{\}$ is causal if the output at the current time n_0 depends only on the current and past inputs; i.e., $y[n_0]$ depends only on $x[n]$ for $n \leq n_0$. (Read defn in book)

Examples: Differencing system: $y[n] = x[n] - x[n-1]$? Causal

Oppenheim/Schaffer
DTSP, 2nd Ed.
PH

Avg system: $y[n] = \frac{1}{2}(x[n] + x[n-1])$? Causal.

Compressor: $y[n] = x[Mn]$ $M > 1$? Non-causal.

Quiz Q 4 c: $y[n] = x[n+1] - x[n-1]$? Non-causal.

Accumulator: $y[n] = \sum_{k=-\infty}^n x[k]$? Causal.

Note: Causal / Non-causal systems play an important role in designing systems with a desired phase response.

- Stability: A system $T\{\}$ is said to be stable in the bounded-input bounded-output (BIBO) sense if all bounded inputs lead to bounded outputs. In other words, if B_x is a finite ^{positive} constant such that $|x[n]| \leq \underline{B_x} < \infty$ for all n , the system is BIBO stable if there exists a B_y such that $|y[n]| \leq B_y < \infty$ for all n .

Example: The accumulator system: $y[n] = \sum_{k=-\infty}^n x[k]$.

Is this system BIBO stable?

- Let $x[n] = u[n] \Rightarrow B_x = 1$ is a valid bound on $x[n]$.
- Is $y[n]$ bounded? No.

$$y[n] = \begin{cases} \boxed{n+1} & n \geq 0 \\ 0 & n < 0 \end{cases} \quad \text{for } x[n] = u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

- We cannot find a constant and finite B_y for this system. $\Rightarrow$ System is not BIBO stable.

Example: $y[n] = \frac{1}{2}(x[n] + x[n-1])$

$\rightarrow B_x$ is a bound on $|x[n]|$ for all n .

$\rightarrow B_y = B_x$.

$$|y[n]| = \left| \frac{1}{2} (x[n] + x[n-1]) \right|$$

$$\leq \frac{1}{2} (B_x + B_x)$$

$$= B_x \quad (\text{for all } n)$$

$\therefore B_y = B_x$ is a valid bound on $|y[n]|$.

- Project using Bookan Board. Dets soon on Google Classroom.